

Topics on Electrical Engineering

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Part I

Magnetic Circuits

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Is it a fact - or have I dreamt it - that, by means of electricity, the world of matter has become a great nerve, vibrating thousands of miles in a breathless point of time?

(Nathaniel Hawthorne)

Chapter 1

Magnetic Circuits and Materials

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The primary objective of these series of lectures is to study devices used in the conversion between **electric** and **mechanical** energy. Of course, as this lecture is about electric drives,¹ emphasis is placed on **electromagnetic rotating machinery**, by means of which the majority of this energy conversion takes place.

¹On paper whereas mostly we will focus more on the machine aspect of drives.

The techniques developed here are generally applicable to a wide range of additional devices including **linear machines**, **actuators**, and **sensors**.

Although **NOT** an electromechanical-energy-conversion device by strictest definitions, the **transformer** is an important component of the overall energy-conversion process and is discussed in Chapter ???. The techniques developed for transformer analysis will form the basis for the ensuing discussion of electric machinery.

Practically, all transformers and electric machinery use **ferromagnetic** material² for shaping and directing the magnetic fields which act as the medium for transferring and converting energy, with Permanent Magnet (PM) materials also being widely used. Without these materials, practical implementations of most familiar electro-mechanical energy-conversion devices would **NOT** be possible.

²A property of certain materials, such as iron, which results in a significant, observable magnetic permeability, and in many cases, a significant magnetic **coercivity**, allowing the material to become a **permanent magnet**. Ferromagnetic materials are noticeably attracted to a magnet, which is a consequence of their substantial magnetic permeability [1].

The ability to analyse and describe systems containing these materials is vital for designing and understanding these devices.

1.1 An Overview on Magnetic Circuits

In this chapter we will develop the necessary tools for analysing of magnetic field systems and will provide a brief introduction to the properties of practical magnetic materials.

1.1.1 Maxwell's Equations

The complete, detailed solution for magnetic fields in most situations of practical engineering interest involves the solution of *Maxwell's equations*³ along with various constitutive relationships which describe material properties. While, in practice, **exact** solutions are often **unattainable**, various simplifying assumptions allow us to have approximate solutions.⁴

³A topic which becomes the center-point in **M.Sc Electrodynamics**.

We start our journey with the assumption that, for the systems treated in this lecture, the frequencies and sizes involved are such that the displacement-current (**D**) term in *Maxwell's equations* can be neglected. This term accounts for magnetic fields being produced in space by time-varying electric fields and is associated with electromagnetic radiation. Neglecting this term results in the magneto-quasistatic form of the relevant Maxwell's equations which relate magnetic fields to the currents which produce them [2]:

⁴Although exact analytical solutions cannot be obtained, numerical solutions are quite common and have become indispensable tools for analysis and design. However, their use contributes little to a fundamental understanding of the principles and basic performance of electric machines and therefore should be used after learning the fundamentals.

$$\oint_C \mathbf{H} \cdot d\mathbf{l} = \int_S \mathbf{J} \cdot d\mathbf{a} \quad (\text{Ampere's law}) \quad (1.1)$$

$$\oint_S \mathbf{B} \cdot d\mathbf{a} = 0 \quad (\text{Gauss's law for magnetic fields}) \quad (1.2)$$

Let's have a look at these two (2) equations and try to understand what they mean;

1. Eq. (1.1) states that the line integral of the tangential component of the magnetic field intensity (**H**) around a closed contour (*C*) is equal to the total current passing through any surface (*S*) linking that contour. From Eq. (1.1) we see that the source of **H** is the current density **J**.
2. Eq. (1.2) states that the magnetic flux density (**B**) is conserved.⁵

⁵i.e., that no net flux enters or leaves a closed surface.

From these equations we see that the magnetic field quantities can be determined **solely** from the instantaneous values of the source currents and time variations of the magnetic fields follow directly from time variations of the sources.

A second simplifying assumption involves the concept of the **magnetic circuit**. The general solution for the magnetic field intensity (**H**) and the magnetic flux density (**B**) in a structure of complex geometry is extremely difficult. However, a three-dimensional field problem can often be reduced to what is essentially a one dimensional circuit equivalent, giving solutions of **within acceptable engineering accuracy** [3].

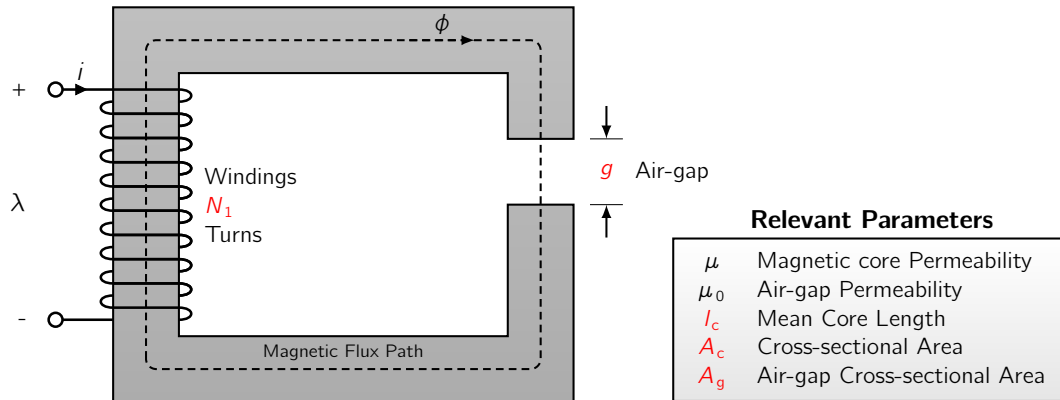


Figure 1.2

Simple magnetic circuit. λ is the winding flux linkage.

A magnetic circuit consists of a structure composed for the most part of **high-permeability** ($\mu \gg 0$) magnetic material. The presence of high-permeability material tends to cause magnetic flux to be **confined** to the paths defined by the structure, similar to currents being confined to the conductors of an electric circuit.

A simple example of a magnetic circuit is shown in **Fig. 1.2**. The core is assumed to be composed of magnetic material whose permeability is much greater than that of the surrounding air ($\mu \gg \mu_0$). The core is of **uniform** cross section and is excited by a winding of N turns carrying a current of i amperes. This winding produces a magnetic field in the core. Due to the high permeability of the magnetic core, an exact solution would show the magnetic flux is confined **almost entirely to the core**, the field lines follow the path defined by the core, and the flux density is **essentially uniform** over a cross section due to the cross-sectional area being uniform.

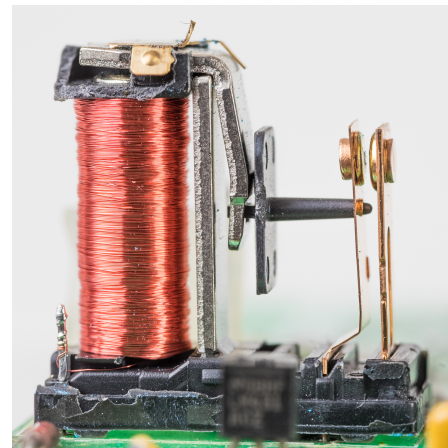


Figure 1.1

An example of a magnetic circuit in action. Relays are used in places where a signal needs to be cut-off at will. These electromechanical relays use magnetism to control the flow of electricity [4].

The magnetic field can be visualised in terms of flux lines which form closed loops interlinked with the winding.

As applied to the magnetic circuit of **Fig. 1.2**, the source of the magnetic field in the core is the ampere-turn product of Ni . In magnetic circuit terminology Ni is defined as the Magneto-motive Force (MMF) ($\mathcal{F}$) acting on the magnetic circuit.

Although **Fig. 1.2** shows only a single coil, transformers and most rotating machines have at least two (2) windings, and Ni therefore needs to be replaced by the **algebraic sum** of the ampere-turns of all the windings.

The magnetic flux (ϕ) crossing a surface ($\mathcal{S}$) is the surface integral of the normal component of

(B). Therefore

$$\phi = \int_S \mathbf{B} \cdot d\mathbf{a} \quad (1.3)$$

In SI units,⁶ the unit of ϕ is the weber (Wb). As we can see, Eq. (1.2) states the net magnetic flux entering or leaving a closed surface, equal to the surface integral of $\mathbf{B}$ over that closed surface, is zero (0).

This is equivalent to saying all flux entering the surface enclosing a volume must leave that volume over some other portion of that surface as magnetic flux lines form closed loops.

These facts can be used to justify the assumption that the magnetic flux density is **uniform** across the cross section of a magnetic circuit such as the core of Fig. 1.2. In this case Eq. (1.3) reduces to the simple *scalar* equation:

$$\phi_c = B_c A_c,$$

where, ϕ_c is the core flux (Wb), B_c is the core flux density ($\text{Wb} \cdot \text{m}^{-2}$), and A_c is the cross-sectional area (m^2). Using Eq. (1.1), the relationship between the MMF acting on a magnetic circuit and the magnetic field intensity in that circuit is:

$$\mathcal{F} = Ni = \oint \mathbf{H} d\mathbf{l}. \quad (1.4)$$

The core dimensions are such that the path length of any flux line is close to the mean core length l_c . As a result, the line integral of Eq. (1.4) becomes simply the scalar product $H_c l_c$ of the magnitude of $\mathbf{H}$ and the mean flux path length l_c . Therefore, the relationship between the MMF and the magnetic field intensity can be written in magnetic circuit terminology as:

$$\mathcal{F} = Ni = H_c l_c$$

where H_c is **average magnitude** of $\mathbf{H}$ in the core.

The direction of H_c in the core can be found from the right-hand rule.⁷

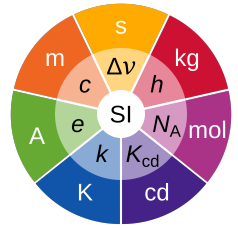
This can be stated in two (2) equivalent but slightly different ways:

- Imagine a current-carrying conductor held in the right hand with the thumb pointing in the direction of current flow. Here, the fingers then point in the direction of the magnetic field created by that current.
- Equivalently, if the coil in Fig. 1.2 is grasped in the right hand with the fingers pointing in the direction of the current, the thumb will point in the direction of the magnetic fields.

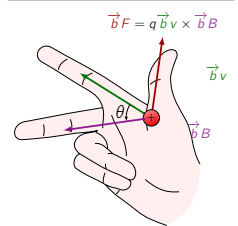
The relationship between the magnetic field intensity $\mathbf{H}$ and the magnetic flux density $\mathbf{B}$ is a property of the material in which the field exists. It is common to assume a **linear relationship**, which means:

$$\mathbf{B} = \mu \mathbf{H} \quad (1.5)$$

where μ is known as the **magnetic permeability**.



⁶The International System of Units, internationally known by the abbreviation SI (from French *Système international d'unités*), is the modern form of the metric system and the world's most widely used system of measurement. It is the only system of measurement with official status in nearly every country in the world, employed in science, technology, industry, and everyday commerce.



⁷A mnemonic, used to define the orientation of axes in three-dimensional space and to determine the direction of the cross product of two vectors, as well as to establish the direction of the force on a current-carrying conductor in a magnetic field.

In SI units, $\mathbf{H}$ is measured in units of $\text{A} \cdot \text{m}^{-1}$, $\mathbf{B}$ is in $\text{Wb} \cdot \text{m}^{-2}$, also known as teslas⁸ (T), and μ is in $\text{Wb} \cdot \text{A}^{-1} \cdot \text{Turn}^{-1} \cdot \text{m}^{-1}$, or equivalently ($\text{H} \cdot \text{m}^{-1}$). In SI units, the permeability of free space is $\mu_0 = 4\pi \times 10^{-7} \text{ H} \cdot \text{m}^{-1}$ and is a **physical constant**.

The permeability of linear magnetic material can be expressed in terms of its **relative permeability**, its value relative to that of free space, or $\mu = \mu_r \mu_0$. Typical values of μ_r range from $2,000 \times 10^3$ to 80,000 for materials used in transformers and rotating machines.

For now we will assume μ_r is a **known** constant, although it actually varies appreciably with the magnitude of the magnetic flux density.

Transformers are wound on closed cores similar to one we have seen in **Fig. 1.2**. However, energy conversion devices which incorporate a moving element must have air gaps in their magnetic circuits which is also shown in **Fig. 1.2**. When the air-gap length (g) is much smaller than the dimensions of the adjacent core faces, the magnetic flux will follow the path defined by the core and the air gap and the techniques of magnetic-circuit analysis can be used.

If the air-gap length were to become excessively large, the flux will be observed to **leak out** of the sides of the air gap and the techniques of magnetic-circuit analysis will no longer be strictly applicable. Therefore, provided the air-gap length g is sufficiently small, the configuration of **Fig. 1.2** can be analysed as a magnetic circuit with two (2) series components both carrying the same flux ϕ :

- a magnetic core of permeability μ , cross-sectional area A_c and mean length l_c , and
- an air gap of permeability μ_0 , cross-sectional area A_g and length g .

In the core, the magnetic field density is approximated to be:

$$B_c = \frac{\phi}{A_c} \quad (1.6)$$

and in the air gap, the value is:

$$B_g = \frac{\phi}{A_g} \quad (1.7)$$

If we apply Eq. (1.4) to aforementioned equations, this gives:

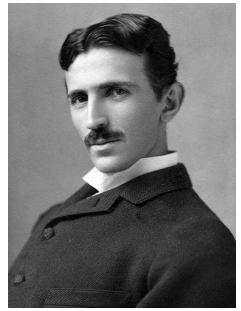
$$\mathcal{F} = H_c l_c + H_g g \quad (1.8)$$

and using the **linear B-H** relationship⁹ of Eq. (1.5) gives

$$\mathcal{F} = \frac{B_c}{\mu} l_c + \frac{B_g}{\mu_0} g \quad (1.9)$$

Here the $\mathcal{F} = Ni$ is the MMF applied to the magnetic circuit. From Eq. (1.8) we can see a portion of the MMF, namely $F_c = H_c l_c$, is required to produce magnetic field in the core whereas the remainder, $F_g = H_g g$ produces magnetic field in the air gap.

For practical applications, B_c and H_c are related **NOT** only to constant permeability μ .



⁸Nikola Tesla
(1856 - 1943)

A Serbian-American engineer, futurist, and inventor. He is known for his contributions to the design of the modern alternating current (AC) electricity supply system. The SI unit is magnetic flux density is named after him.

⁹This is, again, just an assumption of simplification.

In fact, B_c is often a **non-linear**, multi-valued function of H_c . Therefore, although Eq. (1.8) continues to hold, it does not lead directly to a simple expression relating the MMF and the flux densities, such as that of Eq. (1.9).

Instead, the specifics of the nonlinear $B_c - H_c$ relation must be used, either graphically or analytically. However, in many cases, the concept of constant material permeability gives results of acceptable engineering accuracy and is frequently used.

Using both Eq. (1.6) and Eq. (1.7), Eq. (1.9) can be rewritten in terms of the flux ϕ_c as:

$$\mathcal{F} = \phi \left(\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g} \right) \quad (1.10)$$

The terms that multiply the flux in Eq. (1.10), are known as the reluctance¹⁰ ($\mathcal{R}$) of the core and air gap, respectively,

$$\begin{aligned} \mathcal{R}_c &= \frac{l_c}{\mu A_c} \\ \mathcal{R}_g &= \frac{g}{\mu_0 A_g} \end{aligned} \quad (1.11)$$

and therefore we arrive at:

$$\mathcal{F} = \phi(\mathcal{R}_c + \mathcal{R}_g) \quad (1.12)$$

Finally, Eq. (1.12) can be inverted to solve for the flux

$$\begin{aligned} \phi &= \frac{\mathcal{F}}{\mathcal{R}_c + \mathcal{R}_g} \\ \phi &= \frac{\mathcal{F}}{\frac{l_c}{\mu A_c} + \frac{g}{\mu_0 A_g}} \end{aligned} \quad (1.13)$$

In general, for any magnetic circuit of total reluctance ($\mathcal{R}[tot]$), the flux can be found as

$$\phi = \frac{\mathcal{F}}{\mathcal{P}} \quad (1.14)$$

The term which multiplies the MMF is known as the permeance¹¹ ($\mathcal{P}$) and is the inverse of the reluctance. With that in mind, the total permeance of a magnetic circuit is:

$$\mathcal{P}_{tot} = \frac{1}{\mathcal{R}_{tot}}$$

As can be seen, Eq. (1.12) and Eq. (1.13) are analogous to the relationships between the current and voltage in an electric circuit.

This analogy is illustrated in **Fig. 1.3**. In the figure (a) shows an electric circuit in which a voltage V drives a current I through resistors R_1 and R_2 . (b) shows the schematic equivalent representation of the magnetic circuit of **Fig. 1.2**. Here we see that the MMF $\mathcal{F}$, analogous to voltage in the electric circuit, drives a flux ϕ , analogous to the current in the electric circuit, through the combination of the reluctances of the core $\mathcal{R}[c]$ and the air gap $\mathcal{R}[g]$. This analogy between the solution of

¹⁰A concept used in the analysis of magnetic circuits and defined as the ratio of MMF to magnetic flux. It represents the opposition to magnetic flux, and depends on the geometry and composition of an object.

It is analogous to electrical resistance in an electrical circuit in that resistance is a measure of the opposition to the electric current. The definition of magnetic reluctance is analogous to Ohm's law in this respect. However, magnetic flux passing through a reluctance does **NOT** give rise to dissipation of heat as it does for current through a resistance.

¹¹The inverse of reluctance and is a measure of the quantity of magnetic flux for a number of current-turns. A magnetic circuit almost acts as though the flux is conducted, therefore permeance is larger for large cross-sections of a material and smaller for smaller cross section lengths. This concept is analogous to electrical conductance in the electric circuit.

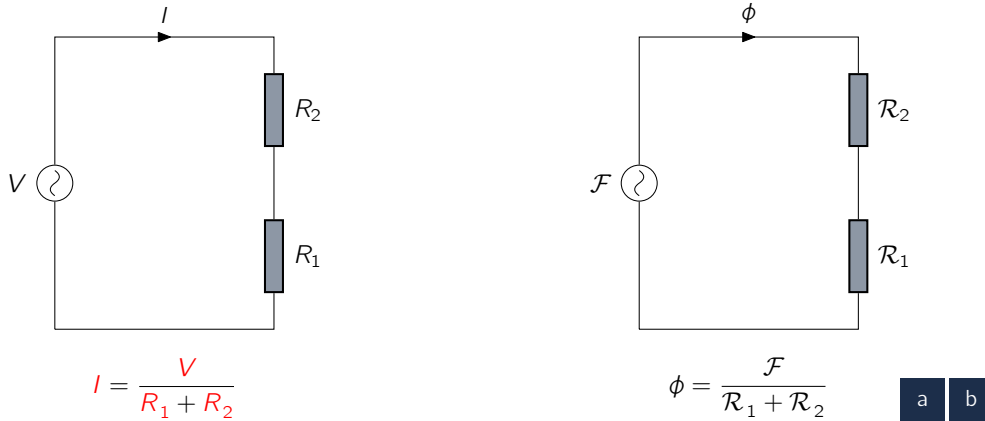


Figure 1.3

A visual comparison to show the conceptual similarities between an electric and magnetic circuit.

electric and magnetic circuits can often be exploited to produce simple solutions for the fluxes in magnetic circuits of considerable complexity.

The fraction of the MMF required to drive flux through each portion of the magnetic circuit, commonly referred to as the **MMF drop** across that portion of the magnetic circuit, varies in proportion to its reluctance.¹²

Consider the magnetic circuit of Fig. 1.2. From Eq. (1.11) we see that high material permeability can result in low core reluctance, which can often be made much smaller than that of the air gap. In this case, the reluctance of the core can be neglected and the flux can be found from Eq. (1.13) in terms of $\mathcal{F}$ and the air-gap properties alone:

$$\phi \approx \frac{\mathcal{F}}{\mathcal{R}_s} = \frac{\mathcal{F}\mu_0 A_g}{g} = Ni \frac{\mu_0 A_g}{g} \quad (1.15)$$

¹² This is directly analogous to the voltage drop across a resistive element in an electric circuit.

Understanding the Assumptions

1.1 Information

While we treat permeability as constant here, it is worth remembering that practical magnetic materials have permeabilities which are not constant but vary with the flux level. However, from Eq. (1.11) to Eq. (1.13) we see that as long as this permeability remains sufficiently large, its variation will not significantly affect the performance of a magnetic circuit in which the dominant reluctance is that of an air gap.

In practical systems, the magnetic field lines “fringe” outward somewhat as they cross the air gap. Provided this fringing effect is **NOT** excessive, the magnetic-circuit concept remains applicable. The effect of these fringing fields is to increase the effective cross-sectional area A_g of the air gap. Various empirical methods have been developed to account for this effect.

A correction for fringing fields in short air gaps can be made by adding the gap length to each of the two dimensions making up its cross-sectional area. For simplicity, we shall ignore fringing effects.

1.2 Flux Linkage, Inductance, and Energy

When a magnetic field varies with time, an electric field is produced in space as per Faraday's¹³ law:

$$\oint_C \mathbf{E} \cdot d\mathbf{s} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{a}$$

Eq. (1.16) states:

The line integral of the electric field intensity ($\mathbf{E}$) around a closed contour ($\mathcal{C}$) is equal to the time rate of change of the magnetic flux linking, passing through, that contour.

In magnetic structures with windings of high electrical conductivity, such as in **Fig. 1.2**, it can be shown, the $\mathbf{E}$ field in the wire is extremely small and can be neglected, so that the Left Hand Side (LHS) of Eq. (1.16) reduces to the negative of the induced voltage (e) at the winding terminals. In addition, the flux on the Right Hand Side (RHS) of Eq. (1.16) is dominated by the core flux ϕ_c . Since the winding, and therefore the contour $\mathcal{C}$, links the core flux N times, Eq. (1.16) reduces to:

$$e = N \frac{d\phi}{dt} = \frac{d\lambda}{dt} \quad (1.16)$$

where λ is the flux linkage of the winding and is defined as:

$$\lambda = N\phi \quad (1.17)$$

Flux linkage is measured in units of Wb (or equivalently Wb · Turn). The symbol ϕ is used to indicate the **instantaneous value** of a time-varying flux. In general the flux linkage of a coil is equal to the **surface integral** of the **normal component of the magnetic flux density** integrated over any surface spanned by that coil.

Direction of the induced voltage (e) is defined by Eq. (1.16) so if the winding terminals were **short-circuited**, a current would flow in such a direction as to **oppose the change of flux linkage**.

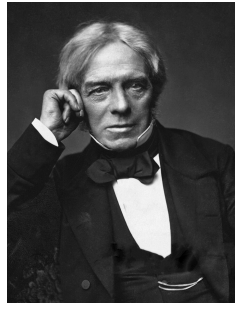
For a magnetic circuit composed of magnetic material of constant magnetic permeability or which includes a dominating air gap, the relationship between λ and i will be **linear** and we can define the inductance L as:

$$L = \frac{\lambda}{i} \quad (1.18)$$

Substitution of Eq. (1.4), Eq. (1.14) and Eq. (1.17) into Eq. (1.18) gives:

$$L = \frac{N^2}{\mathcal{R}_{\text{tot}}}$$

From which we see that the inductance of a winding in a magnetic circuit is proportional to the square of the turns and inversely proportional to the reluctance of the magnetic circuit associated with that winding.



¹³Michael Faraday
(1791 - 1867)

An English chemist and physicist who contributed to the study of electro-chemistry and electromagnetism. His main discoveries include the principles underlying electromagnetic induction, diamagnetism, and electrolysis. Although Faraday received little formal education, as a self-made man, he was one of the most influential scientists in history [5]. It was by his research on the magnetic field around a conductor carrying a direct current Faraday established the concept of the electromagnetic field in physics.

Faraday also established magnetism could affect rays of light and that there was an underlying relationship between the two phenomena. He similarly discovered the principles of electromagnetic induction, diamagnetism, and the laws of electrolysis. His inventions of electromagnetic rotary devices formed the foundation of electric motor technology, and it was largely due to his efforts that electricity became practical for use in technology. The SI unit of capacitance, the farad, is named after him.

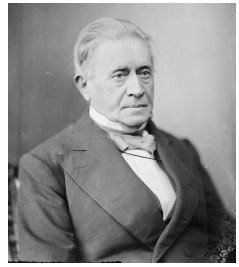
As an example, from Eq. (1.15), under the assumption that the reluctance of the core is negligible as compared to that of the air gap, the inductance of the winding in Fig. 1.2 is equal to

$$L = \frac{N^2}{(g/\mu_0 A_g)} = \frac{N^2 \mu_0 A_g}{g} \quad (1.19)$$

Inductance is measured in henrys¹⁴ (H) or weber-turns per ampere.

Eq. (1.19) shows the dimensional form of expressions for inductance;

Inductance is proportional to the square of the number of turns, to a magnetic permeability, and to a cross-sectional area and is inversely proportional to a length.



¹⁴ Joseph Henry
(1797 - 1878)

An American physicist and inventor who served as the first secretary of the Smithsonian Institution. While building electromagnets, discovered self-inductance. He also discovered mutual inductance independently of Michael Faraday, though Faraday was the first to make the discovery and publish his results and work on the electromagnetic relay was the basis of the practical electrical telegraph. In his honour, the SI unit of inductance is named the henry.

Requirements for Defining Inductance

1.2 Information

It must be emphasised that, strictly speaking, the concept of inductance requires a **linear relationship** between flux and MMF. Therefore, it cannot be rigorously applied in situations where the nonlinear characteristics of magnetic materials, dominate the magnetic system.

However, for many situations of practical interest, the reluctance of the system is dominated by that of an air gap, which is of course linear, and the non-linear effects of the magnetic material can be ignored.

In other cases it may be perfectly acceptable to assume an average value of magnetic permeability for the core material and to calculate a corresponding average inductance which can be used for calculations of reasonable engineering accuracy.

Plotting Permeability

1.1 Exercise

Using Python, plot the inductance of the magnetic circuit given in (TutBook) Ex. ?? as a function of core permeability over the range $100 \leq \mu_r \leq 100000$.

The script for the plot is as follows:

```
#+begin_src python
import numpy as np
import matplotlib.pyplot as plt

mu0 = 4 * np.pi * 1e-7      # (H/m) permeability of free space
lc = 0.3                    # (m) length
Ac = 9e-4                   # (m2) cross-sectional area
Ag = 9e-4                   # (m2) cross-sectional area
g = 5e-4                    # (m) air gap length
N = 500                     # (-) number of turns
Bc = 1                      # (T) Core magnetic field density

Rel_g = (g) / (mu0 * Ag)    # (Ampere-Turns/Weber) air-gap reluctance

mur = np.zeros(1000)       # Relative Permeability empty array for looping

for i in range(len(mur)):
```

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```

mur[i] = 100 + (100000 - 100) * (i - 1)/1000

Rel_c = (lc) / (mur * mu0 * Ac) # (Ampere-Turns/Weber) core reluctance
R_tot = Rel_c + Rel_g          # (Ampere-Turns/Weber) total reluctance

L = N**2 / R_tot               # (H) Inductance

# -- Plotting
plt.xlabel('Core Relative Permeability ($\mu_r$) [H/m]')
plt.ylabel('Inductance [H]')
plt.xlim(0, 0.8e5)
plt.ylim(0, 1)
plt.title("Core Relative Permeability vs. Inductance Graph", color='#c6d0f5')
plt.plot(mur, L, '#ef9f76', linewidth=5)
fig.tight_layout()

```

Please observe the resultant plot. The figure clearly confirms that, for the magnetic circuit of this example, the inductance is quite **insensitive** to relative permeability until the relative permeability drops to on the order of 1000. Therefore, as long as the effective relative permeability of the core is "large", which in this case greater than 1000, any non-linearities in the properties of the core material will have little effect on the terminal properties of the inductor.

Fig. 1.4 shows a magnetic circuit with an air gap and two (2) windings. In this case please observe the MMF acting on the magnetic circuit is given by the **total ampere-turns** acting on the magnetic circuit¹⁵ and that the reference directions for the currents have been chosen to produce flux in the same direction.

¹⁵i.e., the net ampere turns of both windings.

The total MMF is therefore calculated to be:

$$\mathcal{F} = N_1 i_1 + N_2 i_2$$

and from Eq. (1.15), with the reluctance of the core neglected and assuming $A_c = A_g$, the core

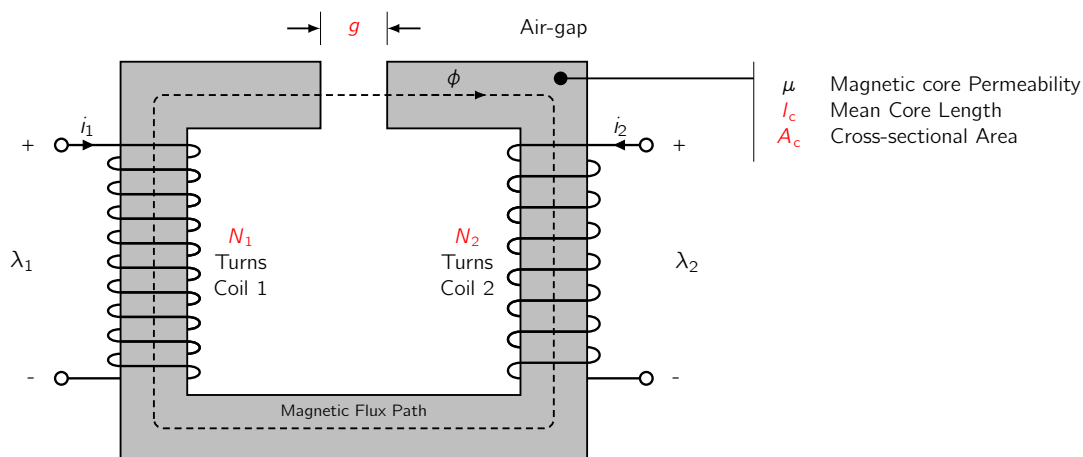


Figure 1.4: An example of a magnetic circuit, excited with two windings.

flux ϕ is:

$$\phi = (N_1 i_1 + N_2 i_2) \frac{\mu_0 A_c}{g} \quad (1.20)$$

In Eq. (1.20), ϕ is the resultant core flux produced by the total MMF of the two windings.

It is this resultant ϕ which determines the operating point of the core material.

If Eq. (1.20) is broken up into individual currents, the resultant flux linkages of coil 1 can be expressed as

$$\lambda_1 = N_1 \phi = N_1^2 \left(\frac{\mu_0 A_c}{g} \right) i_1 + N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right) i_2$$

which can be written in a simpler form as:

$$\lambda_1 = L_{11} i_1 + L_{12} i_2$$

where:

$$L_{11} = N_1^2 \frac{\mu_0 A_c}{g}$$

is the **self-inductance** of Coil 1, and $L_{11} i_1$ is the flux linkage of Coil 1 due to its own current i_1 .

The **mutual inductance** between coils 1 and 2 is:

$$L_{12} = N_1 N_2 \frac{\mu_0 A_c}{g}$$

and $L_{12} i_2$ is the flux linkage of coil 1 due to current i_2 in the other coil. Similarly, the flux linkage of coil 2 is

$$\lambda_2 = N_2 \phi = N_1 N_2 \left(\frac{\mu_0 A_c}{g} \right) i_1 + N_2^2 \left(\frac{\mu_0 A_c}{g} \right) i_2$$

or

$$\lambda_2 = L_{21} i_1 + L_{22} i_2$$

where $L_{21} = L_{12}$ is the mutual inductance and

$$L_{22} = N_2^2 \frac{\mu_0 A_c}{g}$$

is the self-inductance of coil 2.

It is important to note that the resolution of the resultant flux linkages into the components produced by i_1 and i_2 is based on superposition of the individual effects and therefore implies a linear flux-MMF relationship.¹⁶

¹⁶characteristic of materials of constant permeability.

Substitution of Eq. (1.18) in Eq. (1.16) gives:

$$e = \frac{d(Li)}{dt} \quad (1.21)$$

for a magnetic circuit with a single winding. For a static magnetic circuit, the inductance is fixed,¹⁷ and this equation reduces to the familiar circuit-theory form:

¹⁷assuming that material non-linearities do not cause the inductance to vary

$$e = L \frac{di}{dt}$$

However, in electromechanical energy conversion devices, inductances are often time-varying, and Eq. (1.21) must be written as

$$e = L \frac{di}{dt} + i \frac{dL}{dt}$$

Note that in situations with multiple windings, the total flux linkage of each winding must be used in Eq. (1.16) to find the winding-terminal voltage. The power at the terminals of a winding on a magnetic circuit is a measure of the rate of energy flow into the circuit through that particular winding. The **power**, p , is determined from the product of the voltage and the current

$$p = ie = i \frac{d\lambda}{dt} :$$

and its unit is watts (W), or ($\text{J} \cdot \text{s}^{-1}$). Therefore, the change in magnetic stored **energy** ΔW in the magnetic circuit in the time interval t_1 to t_2 is

$$\Delta W = \int_0^{t_2} p dt = \int_0^{\lambda_2} i d\lambda$$

In SI units, the magnetic stored energy W is measured in joules (J). For a single-winding system of constant inductance, the change in magnetic stored energy as the flux level is changed from λ_1 to λ_2 can be written as

$$\Delta W = \int_{\lambda_1}^{\lambda_2} i d\lambda = \int_{\lambda_1}^{\lambda_2} \frac{\lambda}{L} d\lambda = \frac{1}{2L} (\lambda_2^2 - \lambda_1^2) \quad (1.22)$$

The total magnetic stored energy at any given value of λ can be found from setting λ_1 equal to zero:

$$W = \frac{1}{2L} \lambda^2 = \frac{L}{2} i^2$$

The First Electrical Motor

Before their modern versions, motors using electrostatic forces were investigated. The first electric motors were simple electrostatic devices described in experiments by **Andrew Gordon** and **Benjamin Franklin** in the 1740s [7]. However, due to the difficulty of generating the high voltages they required, electrostatic motors were never used for practical purposes.

The invention of the electro-chemical battery by **Alessandro Volta** in 1799 [8] made persistent electric currents possible. **Hans Christian Ørsted** discovered in 1820 that an electric current creates a magnetic field, which can exert a force on a magnet. It only took a few weeks for **André-Marie Ampère** to develop the first formulation of the electromagnetic interaction and present the Ampère's force law, that described the production of mechanical force by the interaction of an electric current and a magnetic field [9].

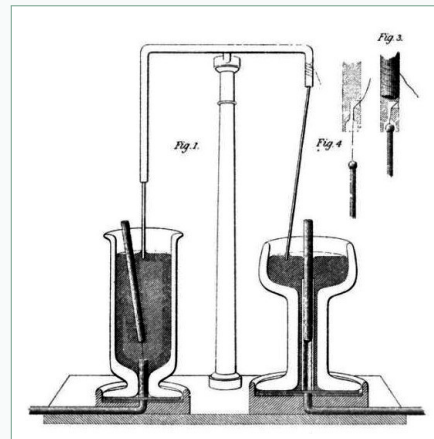


Figure 1.5
Faraday's electromagnetic experiment, 1821, the first demonstration of the conversion of electrical energy into motion [6].

1.3 Information

Continues on next page →

← Continued from previous page

Michael Faraday gave the first demonstration of the effect with a rotary motion in 1821 [10]. A free-hanging wire was dipped into a pool of mercury, on which a PM was placed. When a current was passed through the wire, the wire rotated around the magnet, showing that the current gave rise to a close circular magnetic field around the wire [11].

1.3 Magnetic Material Properties

In the context of electromechanical energy conversion devices, the importance of magnetic materials is twofold. Through their use, it is possible to obtain **large magnetic flux densities** with relatively **low levels of magnetising force**.

As magnetic forces and energy density increase with increasing flux density, this effect plays a large role in the performance of energy-conversion devices.

In addition, magnetic materials can be used to **constrain and direct** magnetic fields in well-defined paths.

For example, in a transformer, they are used to maximise the coupling between the windings as well as to lower the excitation current¹⁸ required for transformer operation. In electric machinery, magnetic materials are used to shape the fields to obtain desired torque-production and electrical terminal characteristics.

Therefore a trained engineer can use magnetic materials to achieve specific desirable device characteristics.

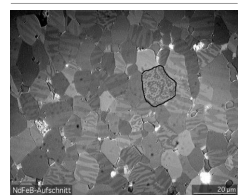
¹⁸The current or amperes required to energise the core. Even with zero load, a transformer will draw a small amount of current due to internal loss. The excitation current is made up of two components. The real component in the form of losses that are commonly referred to as no-load losses. The second form is reactive power measured in KVAR.

1.3.1 Ferromagnetic Materials

These materials are typically composed of iron and alloys of iron with cobalt, tungsten, nickel, aluminium, and other metals, are by far the most common magnetic materials. Although these materials are characterised by a wide range of properties, the basic phenomena responsible for their properties are common to them all.

Ferromagnetic materials are found to be composed of a large number of domains.¹⁹ In a non-magnetised sample of material, the domain magnetic moments are randomly oriented, and the net resulting magnetic flux in the material is zero. When an external magnetising force is applied to this material, the domain magnetic moments tend to **align with the applied magnetic field**. As a result, the domain magnetic moments add to the applied field, producing a much larger value of flux density than would exist due to the magnetising force alone. Therefore, the effective permeability (μ), equal to the ratio of the total magnetic flux density to the applied magnetic-field intensity, is large compared with the permeability of free space (μ_0). As the magnetising force is increased, this behaviour continues until all the magnetic moments are aligned with the applied field at which point they can no longer contribute to increasing the magnetic flux density, and the material is said to be **fully saturated**.

In the absence of an externally applied magnetising force, the domain magnetic moments naturally align along certain directions associated with the **crystal structure of the domain**, known as **axes of easy magnetisation**. Therefore if the applied magnetising force is reduced, the domain magnetic



¹⁹i.e., regions in which the magnetic moments of all the atoms are parallel, giving rise to a net magnetic moment for that domain [12].

Figure 1.6 shows the periodic table of elements. Elements that exhibit ferromagnetism at room temperature are highlighted in blue: Iron (Fe), Cobalt (Co), Nickel (Ni), and Manganese (Mn). The table includes element symbols, atomic numbers, and names. A legend at the bottom left identifies element categories: Alkali Metal (blue), Alkaline Earth Metal (light blue), Metal (grey), Non-metal (green), Halogen (yellow), Noble Gas (orange), and Lanthanide/Actinide (pink).

Figure 1.6

The periodic table of elements. Of all the elements that exist only a few of them have the ability to generate a magnetic field without any external need. The elements which exhibit ferromagnetism has been highlighted.

moments relax to the direction of easy magnetism nearest to that of the applied field. As a result, when the applied field is reduced to zero, although they will tend to relax towards their initial orientation, the magnetic dipole moments will no longer be totally random in their orientation; they will retain a net magnetisation component along the applied field direction. It is this effect which is responsible for the phenomenon known as magnetic hysteresis.

Due to this hysteresis effect, the relationship between **B** and **H** for a ferromagnetic material is **both non-linear and multi-valued**.

In general, material properties cannot be described analytically. They are commonly presented in graphical form as a set of empirically determined curves based on test samples.

The most common curve used to describe a magnetic material is the B-H curve or hysteresis loop which an example is shown in **Fig. 1.7**. Here, only the first quadrant is shown as it is the most important part when designing a machine.

In practice, each curve is obtained while cyclically varying the applied magnetising force between equal positive and negative values of fixed magnitude. Hysteresis causes these curves to be multi-valued. After several cycles the B-H curves form closed loops, which is shown in **Fig. 1.9**.

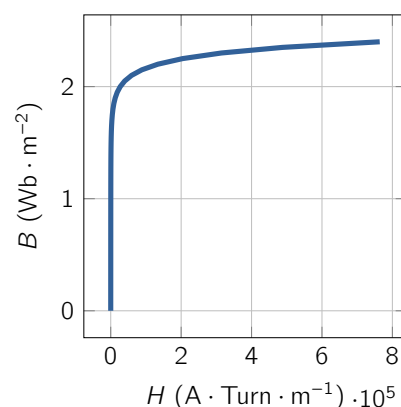


Figure 1.7

An example of a B-H curve showing only the first quadrant.

1.4 AC Excitation

In Alternating Current (AC) power systems, the wave-forms of voltage and flux **closely approximate** sinusoidal functions of time.²⁰ We will now look at the excitation characteristics and losses associated with steady-state AC operation of magnetic materials under such operating conditions. We use as our model a closed-core magnetic circuit,²¹ such as that shown in **Fig. 1.2**. The magnetic path length is l_c , and the cross-sectional area is A_c throughout the length of the core.

²⁰We say here closely as we must take into account environmental factors, such as noise and interference, which distorts the waveform.

²¹This means our circuit has no air gap.

We further assume a sinusoidal variation of the core flux $\phi(t)$, which gives:

$$\phi(t) = \phi_{max} \sin \omega t = A_c B_{max} \sin \omega t$$

where,

- ϕ_{max} is the amplitude of core flux ϕ (Wb),
- B_{max} is the amplitude of flux density B_c (T),
- ω is the angular frequency defined as $\omega = 2\pi f$, and
- f is the frequency (Hz).

From Eq. (1.16), the voltage induced in the N -turn winding is:

$$e(t) = \omega N \phi_{max} \cos \omega t = E_{max} \cos \omega t$$

where

$$E_{max} = \omega N \phi_{max} = E_{max} \cos \omega t$$

In steady-state AC operation, we usually are more interested in the Root-Mean Square (RMS) values of voltages and currents than their instantaneous or maximum values. In general, the RMS value of a periodic function of time, say $f(t)$, of period T is defined as:

$$F_{rms} = \sqrt{\left(\frac{1}{T} \int_0^T f^2(t) dt \right)} \quad (1.23)$$

From Eq. (1.23), the RMS value of a sine wave can be shown to be $1/\sqrt{2}$ times its peak value. Therefore, the RMS value of the induced voltage is:

$$E_{rms} = \frac{2\pi}{\sqrt{2}} f N A_c B_{max} = \sqrt{2} \pi f N A_c B_{max}$$

An excitation current i_ϕ , corresponding to an excitation MMF $Ni_\phi(t)$, is **required** to produce the flux $i_\phi(t)$ in the core.

Because of the non-linear magnetic properties of the core, the excitation current corresponding to a sinusoidal core flux will be **non-sinusoidal**.

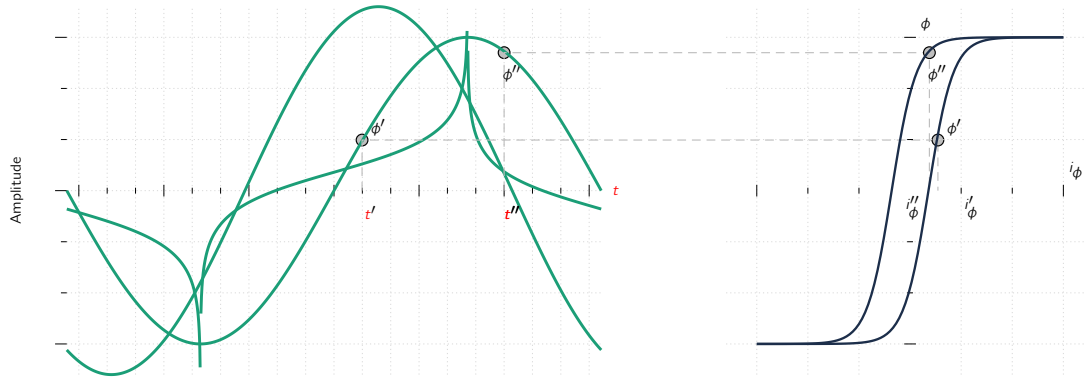


Figure 1.8

Excitation phenomena where left side shows the Voltage, flux, and exciting current and right shows the corresponding hysteresis loop. The current is not to scale.

As B_c and H_c are related to ϕ and i_ϕ by known **geometric constants**, the AC hysteresis loop of **Fig. 1.8** has been drawn in terms of $\phi = B_c A_c$ and is $i_\phi = H_c l_c / N$. The sine waves of induced voltage (e), and flux (ϕ), are shown in the left side of **Fig. 1.8**.

At any given time, the value of i_ϕ corresponding to the given value of flux can be found directly from the hysteresis loop.

As an example, at time t' the flux is ϕ' and the current is i_ϕ' , and at time t'' the corresponding values are ϕ'' and i_ϕ'' . Please observe that since the hysteresis loop is multi-valued, it is necessary to be careful to pick the rising-flux values from the **rising-flux portion** of the hysteresis loop. In the same context, the falling-flux portion of the hysteresis loop must be selected for the falling-flux values.

A second point worth mentioning is, due to the hysteresis loop “flattening out” from the **saturation effects**, the waveform of the exciting current is sharply peaked. Its RMS value $i_{\phi, \text{rms}}$ defined by Eq. (1.23), where T is the period of a cycle. It is related to the corresponding RMS value $H_{c, \text{rms}}$ of H_c by the relationship:

$$i_{\phi, \text{rms}} = \frac{l_c H_{c, \text{rms}}}{N} \quad (1.24)$$

The AC excitation characteristics of core materials are often described in terms of RMS volt-amperes ($\text{V} \cdot \text{A}$) rather than a magnetisation curve relating $\mathbf{B}$ and $\mathbf{H}$. The theory behind this representation can be explained by combining Eq. (1.24) and Eq. (1.25). Therefore, from Eq. (1.24), and Eq. (1.25) the RMS volt-amperes required to excite the core of **Fig. 1.2** to a specified flux density is equal to

$$\begin{aligned} E_{\text{rms}} i_{\phi, \text{rms}} &= \sqrt{2} \pi f N A_c B_{\text{max}} \frac{l_c H_{c, \text{rms}}}{N} \\ &= \sqrt{2} f B_{\text{max}} H_{\text{rms}} (A_c l_c) \end{aligned} \quad (1.25)$$

In Eq. (1.25), the product $A_c l_c$ can be seen to be equal to the volume of the core and hence the RMS exciting volt-amperes required to excite the core with sinusoidal can be seen to be proportional to the frequency of excitation, the core volume and the product of the peak flux density and the RMS magnetic field intensity.

For a magnetic material of mass density ρ_c , the mass of the core is $A_c l_c \rho_c$ and the exciting RMS

voltamperes per unit mass, S_a , can be expressed as

$$S_a = \frac{E_{rms} I_{\phi, rms}}{\text{mass}} = \sqrt{2} \pi f \left(\frac{B_{max} H_{rms}}{\rho_c} \right)$$

Observe that excitation of the material really depends on frequency and B_{max} . As a result, the ac excitation requirements for a magnetic material are often supplied by manufacturers in terms of RMS voltamperes per unit weight as determined by laboratory tests on closed-core samples of the material.

The exciting current supplies the MMF required to produce the core flux and the power input associated with the energy in the magnetic field in the core. Part of this energy is dissipated as losses and results in heating of the core. The rest appears as reactive power associated with energy storage in the magnetic field. This reactive power is not dissipated in the core; it is cyclically supplied and absorbed by the excitation source.

There are two (2) loss mechanisms are associated with time-varying fluxes in magnetic materials.

Eddy Current Losses The first is ohmic ($I^2 R$) heating, associated with induced currents in the core material. From Faraday's law (Eq. (1.16)) we see that time-varying magnetic fields give rise to electric fields. In magnetic materials, these electric fields result in induced currents, commonly referred to as eddy currents, which circulate in the core material and **oppose** changes in flux density in the material. To counteract the corresponding demagnetising effect, the current in the exciting winding must increase.

This causes the resultant "dynamic" B-H loop under ac operation to become somewhat "fatter" than the hysteresis loop for slowly varying conditions, and this effect increases as the excitation frequency is increased.

It is for this reason that the characteristics of electrical steels vary with frequency and therefore manufacturers typically supply characteristics over the expected operating frequency range of a particular electrical steel.

To reduce the effects of eddy currents, magnetic structures are usually built of **thin sheets of laminations** of the magnetic material.²² These laminations, which are **aligned in the direction of the field lines**, are insulated from each other by an oxide layer on their surfaces or by a thin coat of insulating enamel or varnish. This greatly reduces the magnitude of the eddy currents since the layers of insulation interrupt the current paths.

The thinner the laminations, the lower the losses.

In general, eddy-current loss tends to increase as the **square of the excitation frequency** and also as the square of the peak flux density.



²²The primary reason is to reduce the volume in which the current flows. Smaller the volume, smaller the losses [13].

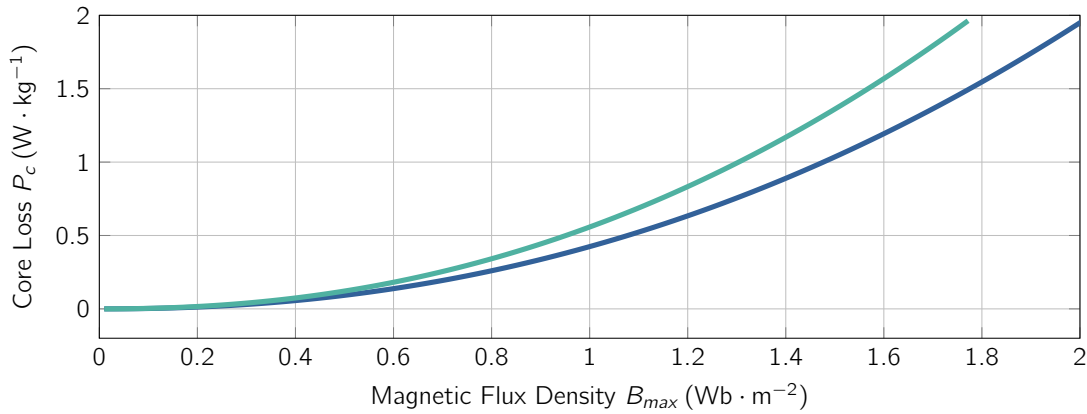


Figure 1.10

Core-loss density of M130-27S, Electric Grain steel under 50 Hz and 60 Hz. The following plot has been generated using the Steinmetz equation.

Hysteresis Losses The second loss mechanism is due to the **hysteretic nature of magnetic material**. In a magnetic circuit similar to **Fig. 1.2**, a time-varying excitation will cause the magnetic material to undergo a cyclic variation described by a hysteresis loop such as that shown in **Fig. 1.9**.

We can use Eq. (1.22) to calculate the energy input W to the magnetic core of **Fig. 1.2** as the material undergoes a single cycle:

$$W = \oint i_\phi d\lambda = \oint \left(\frac{H_c l_c}{N} \right) (A_c N dB_c) = A_c l_c \oint H_c dB_c \quad (1.26)$$

Looking at Eq. (1.26), we see that $A_c l_c$ is the volume of the core and the integral is the area of the AC hysteresis loop, and we see each time the magnetic material undergoes a cycle, there is a net energy input into the material. This energy is required to move around the magnetic dipoles in the material and is dissipated as heat in the material.

Therefore for a given flux level, the corresponding hysteresis losses are proportional to the area of the hysteresis loop and to the total volume of material.

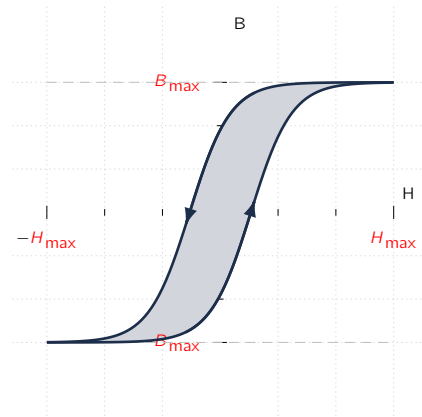


Figure 1.9

Hysteresis loop; hysteresis loss is proportional to the loop area (shaded).

Since there is an energy loss per cycle, hysteresis power loss is proportional to the frequency of the applied excitation.

In general, these losses depend on the metallurgy of the material as well as the flux density and frequency. Information on core loss is typically presented in graphical form. It is plotted in terms of watts per unit weight as a function of flux density; often a family of curves for different frequencies are given. **Fig. 1.10** shows the core loss (P_c) for M130-27S grain-oriented electrical steel at 60 Hz and 50 Hz.

Nearly all transformers and certain sections of electric machines use sheet-steel material that has **highly favourable directions of magnetisation** along which the core loss is low and the permeability

is high. This material is termed **grain-oriented steel**.

Orientation of Atoms

1.4 Information

The reason for this property lies in the atomic structure of a crystal of the silicon-iron alloy, which is a body-centred cube. Each cube has an atom at each corner as well as one in the centre of the cube. In the cube, the easiest axis of magnetisation is the cube edge; the diagonal across the cube face is more difficult, and the diagonal through the cube is the most difficult. By suitable manufacturing techniques most of the crystalline cube edges are aligned in the rolling direction to make it the favourable direction of magnetisation. The behaviour in this direction is superior in core loss and permeability to non-oriented steels in which the crystals are randomly oriented to produce a material with characteristics which are uniform in all directions. As a result, oriented steels can be operated at higher flux densities than the non-oriented grades.

Non-oriented electrical steels are used in applications where the flux does not follow a path which can be oriented with the rolling direction or where low cost is of importance. In these steels the losses are somewhat higher and the permeability is very much lower than in grain-oriented steels.

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Part II

Rotating Machines

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Invention is the most important product of man's creative brain. The ultimate purpose is the complete mastery of mind over the material world, the harnessing of human nature to human needs.

(Nikola Tesla, My Inventions)



Acronyms

		A
AC	An electric current that periodically reverses direction and changes its magnitude continuously with time, in contrast to direct current (DC), which flows only in one direction.	
		L
LHS	Left Hand Side	
		M
MMF	The magnetic pressure which drives magnetic flux in a magnetic circuit, and is the work done to move a unit magnetic pole.	
		P
PM	An object made from a material that is magnetized and creates its own persistent magnetic field	
		R
RHS	Right Hand Side	
RMS	Equivalent direct current (DC) voltage of an alternating current (AC) source.	

